Project Design Phase-II

Solution Requirements

Date	14 February 2026
Team ID	
Project Name	Malaria Detection using Deep Learning
Maximum marks	4

1. Functional Requirements

The functional requirements define the core operations of the system, specifying how users interact with the application and how the system processes inputs to deliver results. These requirements ensure that the system supports image upload, preprocessing, deep learning-based classification, and result generation in an efficient and user-friendly manner. The objective is to automate malaria detection while maintaining accuracy and ease of use for healthcare professionals and end users.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Interface	Web-based image upload system User login and access control
FR-2	Image Upload	Upload blood smear images (JPG/PNG) Validate file format and size
FR-3	Image Processing	Preprocess image (resize & normalize) Convert image for model compatibility
FR-4	Prediction	Use deep learning model for classification Display result (Parasitized / Uninfected)
FR-5	Result Display	Show prediction with confidence score Provide user-friendly output
FR-6	Data Storage (Optional)	Store user history and results Secure database integration

Table 1.1 Functional Requirements

2. Non-functional Requirements

The non-functional requirements define the quality attributes of the system, ensuring that it operates efficiently, securely, and reliably under various conditions. These requirements focus on system performance, usability, security, and scalability, which enhance user experience and long-term sustainability. By addressing non-functional aspects, the system ensures fast response times, data protection, and adaptability to future enhancements, making it a robust solution for AI-based malaria detection.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple and user-friendly web interface
NFR-2	Security	Secure file upload and input validation
NFR-3	Reliability	Consistent and accurate predictions
NFR-4	Performance	Fast image processing and response time
NFR-5	Availability	Web-based system accessible anytime
NFR-6	Scalability	Supports future expansion and features

Table 1.2 Non-Functional Requirements